Do Now · retrieval

Karnaugh maps 2: simplify to gates

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

In a Karnaugh-map pair, A stays 0 and B changes between 0 and 1. What is the simplified term?

Q2

Show the array after one complete left-to-right pass of ascending bubble sort on [8, 3, 6, 2] .

Q3

C#: int[,,] scores = new int[2, 3, 4]; How many values can it store? What does scores.GetLength(2) return?

Q4

Add 01000000 0011 and 01000000 0010 . Format: 8-bit two’s-complement fractional mantissa + 4-bit two’s-complement exponent . Show exponent alignment and the final normalised result.

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Find the 1s — Look at the 2×2 map on the left. Three squares contain 1 . We need to cover them in groups of neighbouring 1s. — Use rectangles containing 1, 2, 4, 8 or 16 cells. Include only 1s, and choose the largest useful groups. Diagonal cells are not neighbours.

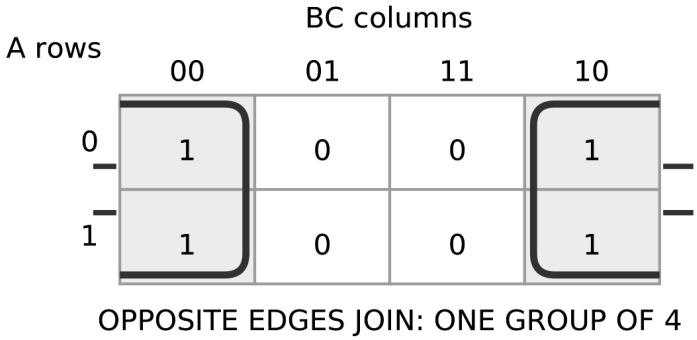
Core idea 2

A square may belong to two groups — This is overlap . It is allowed when it makes the expression simpler.

Core idea 3

Keep the inputs that stay the same — Bottom pair: A stays 1; B changes. Keep A. — Right pair: B stays 1; A changes. Keep B.

Key visual · complete concept



OPPOSITE EDGES JOIN: ONE GROUP OF 4

Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Core idea 4

The outside columns form one group — BC runs 00, 01, 11, 10 . The first and last columns are neighbours: only B changes between 00 and 10 . — Imagine rolling the map so its left and right edges meet. The four 1s make a 2 × 2 group ; the middle 0s are outside it.

Core idea 5

Compare all four cells — A changes between the two rows → remove A. — B changes between columns 00 and 10 → remove B.

Vocabulary · use the correct technical terms

efficient group — A group of 1, 2, 4, 8 or 16 neighbouring 1s.

overlap — The same 1 can appear in two different groups.

wraparound — The left/right edges touch, and the top/bottom edges touch.

literal — An input letter, such as A, or its NOT, ¬A.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

How do two groups become a circuit?

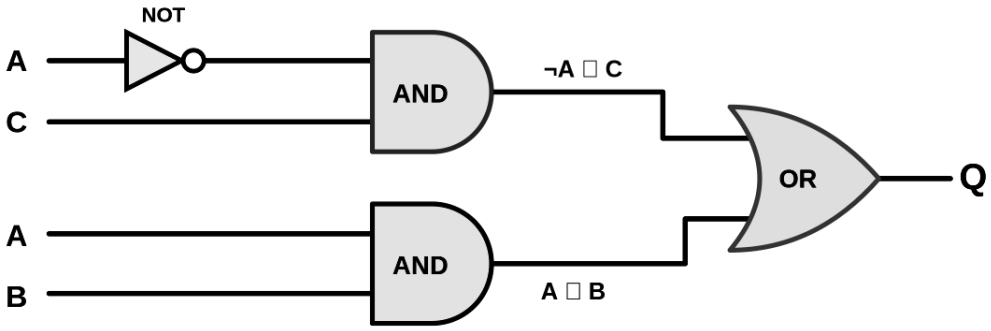
Follow the worked method

1. Find the two pairs of 1s — ORANGE · TOP: circle columns 01 and 11 where A = 0. — BLUE · BOTTOM: circle columns 11 and 10 where A = 1. — Follow the same coloured groups on the map on the left. Each pair becomes one term.

2. Keep A and B — A = 1 in both squares, so keep A . B = 1 in both, so keep B . C changes , so remove C.

3. Draw the actual logic gates — Trace it: A goes through NOT for the orange pair; A also goes directly to the blue pair. Each pair feeds an AND gate. OR the two AND outputs to get Q.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

1. Find the 1s

Circle the two 1s. Which two inputs always equal 1? Which input changes?

Guided K-map · mark the pair

A rows	BC columns: 00	01	11	10
0	0	0	0	0
1	0	1	1	0

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Find one pair
Circle the largest valid groups of 1s.
Compare the inputs that remain constant in each group.
Simplify to the Boolean expression Q.
Draw the logic gates and connections.

Question 1 · original source diagram

A rows \ BC columns	00	01	11	10
0	1	1	0	0
1	0	0	0	0

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 2

Two groups in a bigger map
Circle the largest valid groups of 1s.
Compare the inputs that remain constant in each group.
Simplify to the Boolean expression P.
Draw the logic gates and connections.

Question 2 · original source diagram

AB rows \ CD columns	00	01	11	10
00	0	0	1	1
01	0	0	1	1
11	1	0	0	1
10	1	0	0	1

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

A K-map simplification is valid only if the simplified expression gives the same output for every possible input combination.

Use key vocabulary from memory

1. Explain efficient group in your own words.

2. Explain overlap in your own words.

3. Explain wraparound in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.